





- 1 A thin cylindrical bar magnet of length  $L$  and cross-sectional area  $A$  is attached to a block. An identical magnet is attached to a trolley, as shown in Fig. 1.1.

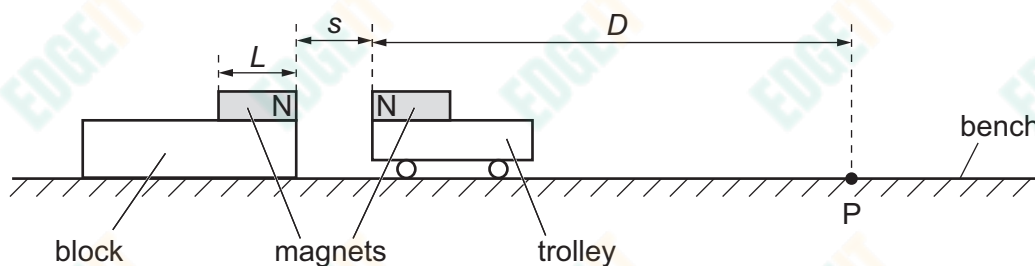


Fig. 1.1

The trolley is held so that the separation of the N poles of the two magnets is  $s$ .

Point P is a distance  $D$  from the N pole of the magnet on the stationary trolley.

The trolley is released. The speed  $v$  of the trolley at point P is determined using one light gate.

It is suggested that  $v$  is related to  $s$  by the relationship

$$\frac{mv^2}{2D} = \frac{KA^2B^2L^2}{s^4} - Q$$

where  $B$  is the magnetic flux density at the N pole of one of the magnets,  $m$  is the mass of the trolley, and  $K$  and  $Q$  are constants.

Plan a laboratory experiment to test the relationship between  $v$  and  $s$ .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for  $K$  and  $Q$ .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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- 2 A student investigates an electrical circuit. A power supply of electromotive force (e.m.f.)  $E_s$  and negligible internal resistance is connected in series to three resistors, each of resistance  $Z$ .

A cell, an ammeter and a resistor of resistance  $R$  are connected in parallel across one of these resistors, as shown in Fig. 2.1.

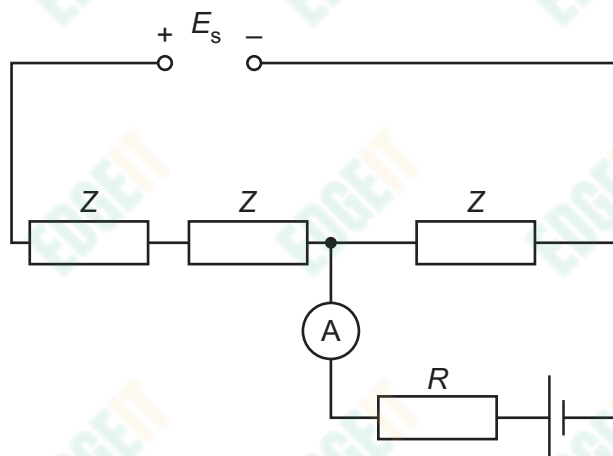


Fig. 2.1

The current  $I$  is measured by the ammeter for different values of  $R$ .

It is suggested that  $I$  and  $R$  are related by the equation

$$3E - E_s = I(3R + 2Z)$$

where  $E$  is the e.m.f. of the cell.

- (a) A graph is plotted of  $\frac{1}{I}$  on the  $y$ -axis against  $R$  on the  $x$ -axis.

Determine expressions for the gradient and  $y$ -intercept.

gradient = .....

$y$ -intercept = .....

[1]





(b) Values of  $R$  and  $I$  are given in Table 2.1.

Table 2.1

| $R/\text{k}\Omega$ | $I/\mu\text{A}$ | $\frac{1}{I}/\text{A}^{-1}$ |
|--------------------|-----------------|-----------------------------|
| 1.50               | $194 \pm 2$     |                             |
| 1.75               | $180 \pm 2$     |                             |
| 1.92               | $172 \pm 2$     |                             |
| 2.22               | $160 \pm 2$     |                             |
| 2.48               | $150 \pm 2$     |                             |
| 2.72               | $144 \pm 2$     |                             |

Calculate and record values of  $\frac{1}{I}/\text{A}^{-1}$  in Table 2.1.

Include the absolute uncertainties in  $\frac{1}{I}$ .

[2]

(c) (i) Plot a graph of  $\frac{1}{I}/\text{A}^{-1}$  against  $R/\text{k}\Omega$ . Include error bars for  $\frac{1}{I}$ .

[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

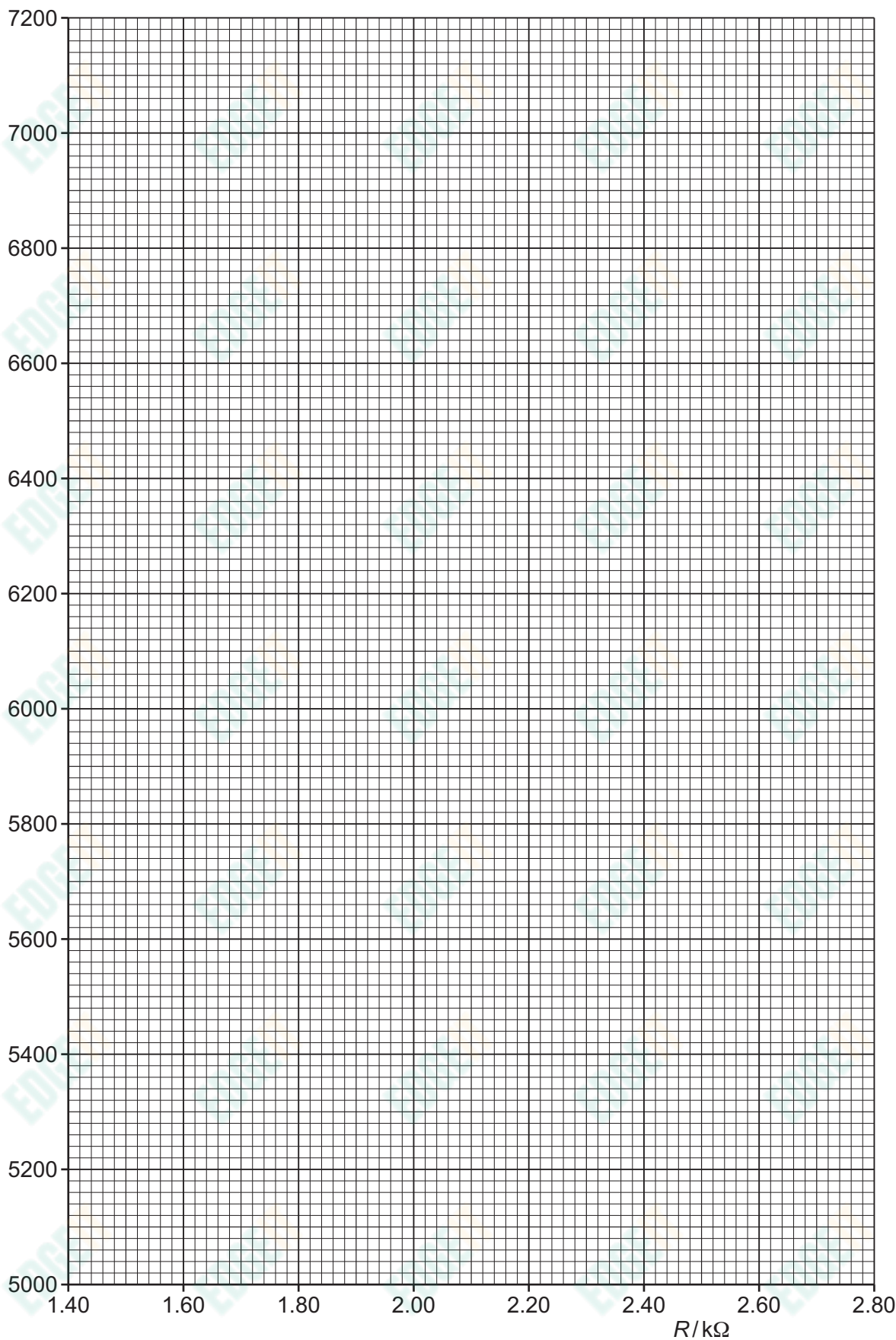
(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ..... [2]





$\frac{1}{I} / \text{A}^{-1}$





- (iv) Determine the  $y$ -intercept of the line of best fit. Include the absolute uncertainty in your answer.

$y$ -intercept = ..... [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of  $E$  and  $Z$ . Include appropriate units.

Data:  $E_s = (2.20 \pm 0.05)V$

$E =$  .....

$Z =$  ..... [2]

- (ii) Determine the absolute uncertainty in  $E$ .

absolute uncertainty in  $E =$  ..... [1]

- (e) The experiment is repeated. Determine the resistance  $R$  that gives a value of  $I$  of  $250\mu A$ .

$R =$  .....  $\Omega$  [1]

[Total: 15]

